

STRESS PROPAGATION THROUGH JOINTED BEAM SYSTEMS USING FINITE ELEMENT THEORY

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Summary—The propagation of disturbances through connected beams is considered using finite element theory: the Timoshenko beam model is considered since it exhibits the correct propagation behaviour in single beams. The joint is examined and three joint models are introduced into the model, namely the rigid block model, the filament model and finally a flexible joint model. The numerical results so obtained compare favourably with experimental results produced at UMIST.

NOTATION

A	cross-sectional area of beam
$[C_1], [C_2], [C_3], [C_4], [C_5]$	coefficient matrices
E	Young's modulus
G	shear modulus
I	second moment of area about centroid
I_B	polar moment of inertia of rigid block
L	Lagrangian
M_B	mass of rigid block
$\{Q\}, \{R_1\}, \{R_2\}, \{R_3\}$	column matrices
T	transformation matrix
l	length of beam
m	number of beams connected together at joint
n	number of finite elements in a beam
q	$\{u \ v \ \theta\}^T$
t	time
u	axial displacement
v	transverse displacement
x, ξ	axial co-ordinates
δ	variational operator
Δx	length of finite element
Δt	time increment
∇q	$\{\partial/\partial u \ \partial/\partial v \ \partial/\partial \theta\}$
λ	scale factor
ν	Poisson's ratio
ϕ	transformation angle
ρ	density
θ	rotation of plane section of beam
$(\dot{})$	denotes differential with respect to time
$(\prime)$	denotes differential with respect to x
()	denotes global co-ordinates

INTRODUCTION

The dynamics of beams and in particular beam structures has long been recognised as a problem of practical importance. Beam models are developed in general from the various kinematics assumed and the behaviour of these models will be dictated by the situations in which these beams are placed.

For stress wave problems one automatically excludes the Euler-Bernoulli beam since it does not exhibit finite propagation velocities for impulsive type loads. The other beam models in the group, namely the shear, Rayleigh and Timoshenko beams, all have finite propagation velocities. The shear beam propagates disturbances, without bending at the shear propagation speed; the Rayleigh and Timoshenko beam propagates disturbance at the flexural propagation speed and in the latter case also at the shear propagation speed. Generally beam systems do not exhibit large shear

deformation and it would appear that the Rayleigh beam is optimum. However for jointed systems in which various beams are connected, it is probable that there is shear deformation in the vicinity of the joint.

This paper considers the propagation of disturbances through the joint of connected beams and for the analysis Timoshenko beam theory is used; to complete the analysis finite element theory is introduced and the numerical results obtained are compared with experimental results obtained at UMIST.

Previous investigators[1-3] have considered this problem and this investigation complements these previous investigations. Mandel *et al.*[1] considered the Timoshenko beam theory to model right angled beam systems. The joint is assumed to be a rigid block and the equations were then solved by the method of characteristics. Lee and Kolsky[2] also used the Timoshenko model and used a Fourier analysis to predict the shape of the reflected and transmitted longitudinal and flexural pulses developed through a right angled system. They did not however consider shear generated by the joint. Atkins and Hunter[3] considered the same system using the Euler-Bernoulli beam and also considered the rigid block approximation for the joint region.

In this paper various joint models are examined and various beam configurations are considered, Fig. 1. The solutions being numerical rather than analytical are displayed in graphical form and in general compare favourably with the experimental results.

BEAM THEORIES

Shear, Euler-Bernoulli, Rayleigh and Timoshenko beam equations are often derived by differential calculus. However, it is often more convenient to derive these equations by means of variational principles and in particular Hamilton's principle[4].

Hamilton's principle for a conservative system states that of all possible motions of an arbitrary mechanical system the actual motion of that system minimises the integral, with respect to time of the difference between the kinetic and potential energy, provided the initial and final configurations of the system are prescribed.

In mathematical terms Hamilton's principle can be written as,

$$\delta \int_{t_1}^{t_2} L \, dt = 0$$

where L is the Lagrangian and is the difference between the kinetic and potential energy of the system. It should also be noted that the principle can be used practically with simplified approximate descriptions of

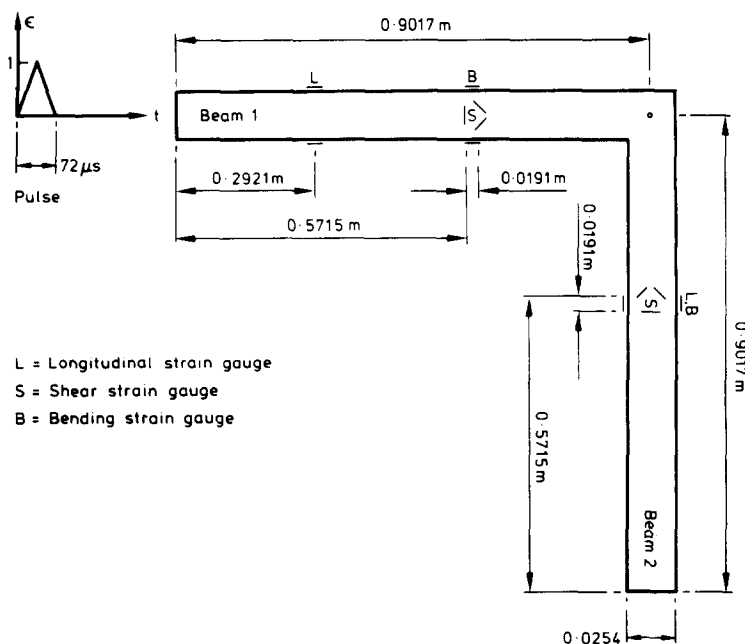


FIG. 1(a). Right angled beam system.

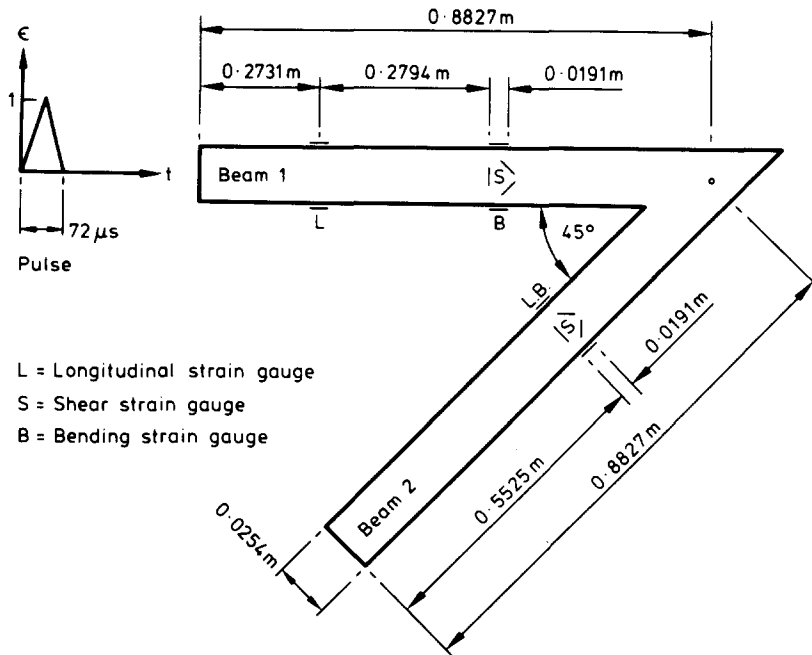


FIG. 1(b). 45° angled beam system.

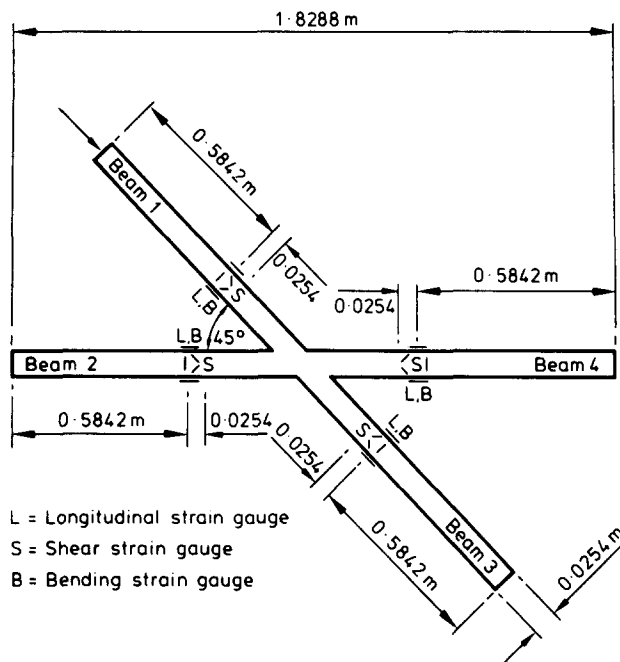


FIG. 1(c). Angled cruciform beam system.

beam behaviour and that the variation then produces the optimal equations for the approximation employed.

Of the four beam theories mentioned above, Timoshenko theory is the most complex. The remaining three theories may be shown to be mathematical approximations to Timoshenko theory.

Before deriving Timoshenko beam theory, it is necessary to consider the basic assumptions governing the analysis:

- (1) The beam is made of material which is homogeneous, isotropic, linearly elastic, has density ρ and elastic moduli E and G and Poisson's ratio ν where $G = E/2(1 + \nu)$.
- (2) Plane sections remain plane but are allowed to rotate relative to the median.
- (3) Transverse shear strain is constant over the cross-section of the beam.
- (4) The effects of shear deformation and rotary inertia are not negligible.

Fig. 2 shows a typical Timoshenko beam element. The Lagrangian for the whole beam may be expressed in terms of three generalised co-ordinates, u , v and θ , so that

$$L = \int_0^l \frac{1}{2} \{ (\rho A \dot{u}^2 + \rho A \dot{v}^2 + \rho I \dot{\theta}^2) - (E A u'^2 + G A (v' - \theta)^2 + E I \theta'^2) \} dx.$$

Inserting the Lagrangian into Hamilton's principle and performing the variation results in the following equations.

$$\rho \ddot{u} - E u'' = 0 \tag{1}$$

$$\rho \ddot{v} - G (v'' - \theta') = 0 \tag{2}$$

$$\rho I \ddot{\theta} - E I \theta'' - G A (v' - \theta) = 0 \tag{3}$$

with the following boundary conditions at $x = 0$ and $x = l$

$$\begin{aligned} u' = 0 \quad \text{or} \quad u \text{ prescribed} \\ v' - \theta = 0 \quad \text{or} \quad v \quad \quad \quad \text{''} \\ \theta' = 0 \quad \text{or} \quad \theta. \quad \quad \quad \text{''} \end{aligned}$$

Equation (1) is simply the one-dimensional wave equation describing axial wave propagation. Equations (2) and (3) are the Timoshenko beam equations which may be decoupled by eliminating terms in θ to give the Timoshenko beam equation in more conventional form,

$$E I v'''' + \rho A \ddot{v} - \rho I \left(1 + \frac{E}{G} \right) \ddot{v}'' + \frac{\rho^2 I}{G} \ddot{v} = 0.$$

To obtain the Shear, Euler–Bernoulli and Rayleigh beam equations it is only necessary to equate relevant terms in the Timoshenko Lagrangian to zero. This is summarised in the table below.

TABLE 1.

Beam theory	Assumptions	Equations	Boundary conditions $x = 0$ and $x = l$
Rayleigh	No shear deformation, i.e. $v' - \theta = 0$	$\rho \ddot{u} - E u'' = 0$	$\begin{matrix} u' = 0 & \text{or} & u & \text{Prescribed} \\ \rho \ddot{v} - E v''' = 0 & \text{or} & v & \text{''} \\ v'' = 0 & \text{or} & v' & \text{''} \end{matrix}$
		$\rho A \ddot{v} - \rho I \ddot{v}'' + E I v'''' = 0$	
Euler–Bernoulli	No shear deformation and no rotary inertia i.e. $v' - \theta = 0$ $\frac{1}{2} \rho I \dot{\theta}^2 = 0$	$\rho \ddot{u} - E u'' = 0$	$\begin{matrix} u' = 0 & \text{or} & u & \text{''} \\ v'' = 0 & \text{or} & v' & \text{''} \\ v''' = 0 & \text{or} & v & \text{''} \end{matrix}$
		$\rho A \ddot{v} + E I v'''' = 0$	
Shear	No rotation, i.e. $\theta = 0$ $\dot{\theta} = 0$ $\theta' = 0$	$\rho \ddot{u} - E u'' = 0$	$\begin{matrix} u' = 0 & \text{or} & u & \text{''} \\ v' = 0 & \text{or} & v & \text{''} \end{matrix}$
		$\rho \ddot{v} - G v'' = 0$	

FINITE ELEMENT FORMULATION

An alternative to solving the Timoshenko partial differential equations of motion is to use a finite element formulation.

The use of the latter method becomes increasingly advantageous when considering multiple beam systems. However, for the present, consider the Timoshenko Lagrangian of a single beam subdivided into n (finite) elements each of length Δx , as shown in Fig. 3.

The Lagrangian may be written as

$$L = \sum_{i=1}^n L_i = \sum_{i=1}^n \int_0^{\Delta x} \frac{1}{2} \{ (\rho A \dot{u}^2 + \rho A \dot{v}^2 + \rho I \dot{\theta}^2) - (E A u'^2 + G A (v' - \theta)^2 + E I \theta'^2) \} dx.$$

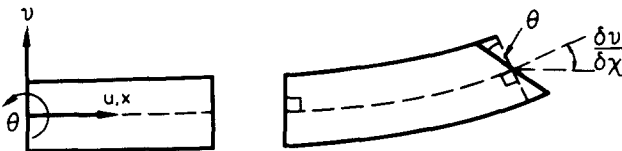


FIG. 2. Timoshenko beam element.

Considering the i th element, Fig. 4, the generalised co-ordinates u , v and θ may be written in terms of the nodal co-ordinates.

$$\left. \begin{aligned} u &= u_i(1 - \xi/\Delta x) + u_{i+1}\xi/\Delta x \\ v &= v_i(1 - \xi/\Delta x) + v_{i+1}\xi/\Delta x \\ \theta &= \theta_i(1 - \xi/\Delta x) + \theta_{i+1}\xi/\Delta x \end{aligned} \right\} 0 \leq \xi \leq \Delta x$$

Inserting the above interpolation functions into the functional, integrating and performing the variation yields,

$$\left. \begin{aligned} \frac{\partial}{\partial t} \left(\frac{\partial L}{\partial \dot{u}_i} \right) - \frac{\partial L}{\partial u_i} &= 0 \quad \text{or } u_i \text{ prescribed} \\ \frac{\partial}{\partial t} \left(\frac{\partial L}{\partial \dot{v}_i} \right) - \frac{\partial L}{\partial v_i} &= 0 \quad \text{or } v_i \text{ prescribed} \\ \frac{\partial}{\partial t} \left(\frac{\partial L}{\partial \dot{\theta}_i} \right) - \frac{\partial L}{\partial \theta_i} &= 0 \quad \text{or } \theta_i \text{ prescribed} \end{aligned} \right\} i = 1, \dots, (n+1).$$

This set may be written more compactly as

$$\left\{ \frac{\partial}{\partial t} (\nabla_{\dot{q}_i} L) - \nabla_{q_i} L \right\} \{\delta q_i\} = 0, \quad i = 1, \dots, (n+1) \quad (4)$$

where

$$\delta q_i = \{\delta u_i \delta v_i \delta \theta_i\}^T,$$

and the operators $\nabla_{\dot{q}_i}$ and ∇_{q_i} are defined by the following

$$\nabla_{\dot{q}_i} = \left\{ \frac{\partial}{\partial \dot{u}_i} \quad \frac{\partial}{\partial \dot{v}_i} \quad \frac{\partial}{\partial \dot{\theta}_i} \right\},$$

$$\nabla_{q_i} = \left\{ \frac{\partial}{\partial u_i} \quad \frac{\partial}{\partial v_i} \quad \frac{\partial}{\partial \theta_i} \right\}.$$

Equation (4), once all the partial differentials have been evaluated, may in turn be expressed in matrix form,

$$[C_1]\{\ddot{Q}\} + [C_2]\{\dot{Q}\} = \{R_1\}, \quad (5)$$

where

$$\{Q\} = \{u_1 \dots u_{n+1}, v_1 \dots v_{n+1}, \theta_1 \dots \theta_{n+1}\}^T.$$

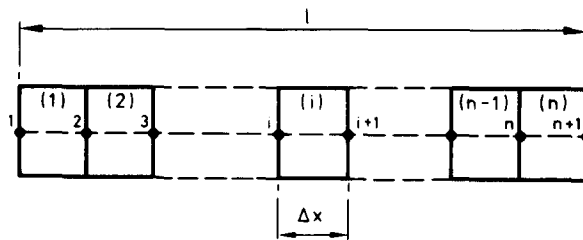


FIG. 3. Finite elements.

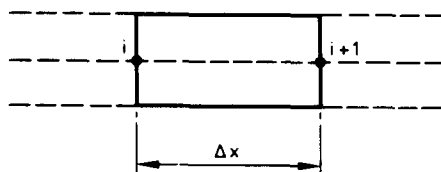


FIG. 4. i th element.

Equation (5) is set of simultaneous second order ordinary differential equations and can be solved by incorporating the following finite difference approximations,

$$\begin{aligned}\{\ddot{Q}\} &\approx \{\{Q\}^{t+\Delta t} - 2\{Q\}^t + \{Q\}^{t-\Delta t}\} \frac{1}{\Delta t^2}, \\ \{Q\} &\approx \{\{Q\}^{t+\Delta t} + 2\{Q\}^t + \{Q\}^{t-\Delta t}\} \frac{1}{4}\end{aligned}$$

where Δt is some suitable time step. These relationships are substituted into equation (5) to give

$$[C_3]\{Q\}^{t+\Delta t} = [C_4]\{Q\}^t + [C_5]\{Q\}^{t-\Delta t} + \{R_2\} = \{R_3\}.$$

Since the only unknown is the $\{Q\}^{t+\Delta t}$ matrix,

then

$$\{Q\}^{t+\Delta t} = [C_3]^{-1}\{R_3\}.$$

MULTIPLE BEAM SYSTEMS

Consider a number of beams connected together to form a single joint as shown in Fig. 5. The Lagrangian for this system may be written as,

$$L = \sum_{j=1}^m \sum_{i=1}^n L_{ij} \quad (6)$$

where n is the number of elements in each beam and m is the number of beams. Three types of joint model will be considered. These are the filament, rigid block and one-dimensional triangular finite element models.

FILAMENT MODEL

The filament approximation to the joint is a direct extension to the analysis presented so far but, as may be observed from Fig. 6, does not accurately represent the mass distribution of the joint. Substitution of the interpolation functions into equation (6) integrating and performing the variation yields,

$$\left\{ \frac{\partial}{\partial t} (\nabla \dot{q}_l / L) - \nabla q_l / L \right\} \{\delta q_l\} = 0, \quad \begin{matrix} i = 1, \dots, n \\ j = 1, \dots, m \end{matrix} \quad (7)$$

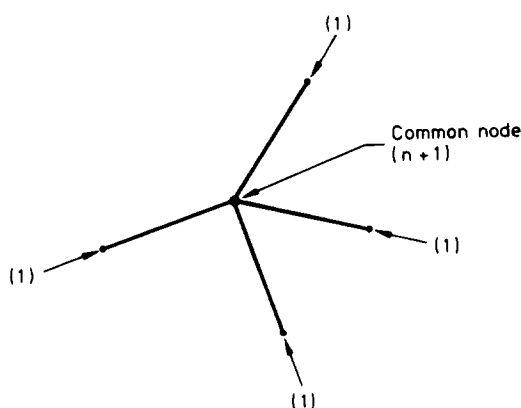


FIG. 5. Multiple beam system.

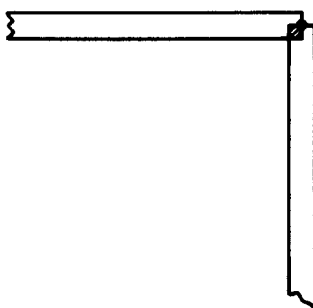


FIG. 6. Filament approximation.

and

$$\sum_{j=1}^m \left\{ \frac{\partial}{\partial t} (\nabla \dot{q}_{n+1}^j L) - \nabla q_{n+1}^j L \right\} \{\delta q_{n+1}^j\} = 0. \quad (8)$$

Equation (8) is the variational equation for the joint and occurs as a summation because the variations at the joint of each beam are not independent. By choosing a global set of co-ordinates at the joint, (Fig. 7), and the transformation equation,

$$\{q_{n+1}^j\} = [T_j] \{\bar{q}\},$$

where

$$[T_j] = \begin{Bmatrix} \cos \phi_j & \sin \phi_j & 0 \\ -\sin \phi_j & \cos \phi_j & 0 \\ 0 & 0 & 1 \end{Bmatrix},$$

then equation (8) becomes

$$\left\{ \sum_{j=1}^m \left\{ \frac{\partial}{\partial t} (\nabla \dot{q}_{n+1}^j L) - \nabla q_{n+1}^j L \right\} [T_j] \right\} \{\delta \bar{q}\} = 0 \quad (9)$$

and now the variations may be taken one at a time and the nodal displacements determined by considering equations (7) and (9) as a simultaneous set.

RIGID BLOCK MODEL

In the previous joint analysis the mass distribution is incorrect. The simplest way to model the correct mass distribution is to approximate the joint by a rigid block, as shown in Fig. 8. Atkins and Hunter[3] used this assumption in their work, although for quite different reasons. They used Euler-Bernoulli theory together with a rigid block joint to predict the axial strain present in the second leg of the system when the first leg is subjected to an initial compressive pulse.

The Lagrangian for the block may be written in terms of global co-ordinates,

$$L_B = \frac{M_B}{2} (\dot{\tilde{u}} + \dot{\tilde{v}})^2 + \frac{I_B}{2} \dot{\tilde{\theta}}^2,$$

where M_B = mass of the block; I_B = second moment of inertia of the block about its centroid. The equation at the joint may now be written as,

$$\left\{ \frac{\partial}{\partial t} (\nabla \dot{q} L_B) + \sum_{j=1}^m \left\{ \frac{\partial}{\partial t} (\nabla \dot{q}_{n+1}^j L) - \nabla q_{n+1}^j L \right\} [T_j] \right\} \{\delta \bar{q}\} = 0. \quad (10)$$

The rigid block approximation, whilst modeling the correct mass distribution, does not allow strain to exist within the joint and therefore any disturbance arriving at one side of the joint is propagated across it with an infinite velocity.

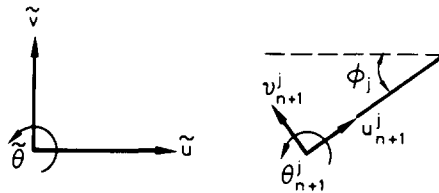


FIG. 7. Global and local co-ordinate systems.

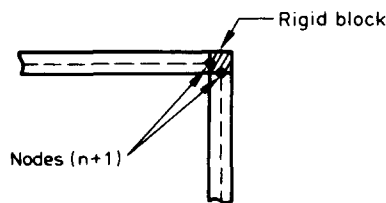


FIG. 8. Rigid block approximation.

TRIANGULAR FINITE ELEMENT APPROXIMATION TO THE JOINT

An alternative to the rigid block and filament models is to consider the joint as being composed of one-dimensional triangular finite elements, as shown in Figs. 9 and 10. The interpolation functions for the triangular element may be written as,

$$u = u_n(1 - 2\xi/\Delta x) + u_{n+1}2\xi/\Delta x,$$

$$v = v_n(1 - 2\xi/\Delta x) + v_{n+1}2\xi/\Delta x,$$

$$\theta = \theta_n(1 - 2\xi/\Delta x) + \theta_{n+1}2\xi/\Delta x.$$

Unfortunately, the two-noded triangular element analysis is restricted to the two beam system.

The equation at the joint for a two beam system now becomes,

$$\left\{ \sum_{j=1}^2 \left\{ \frac{\partial}{\partial t} (\nabla \dot{q}_{n+1}^j L) - \nabla \dot{q}_{n+1}^j L \right\} [T_j] \right\} \{\delta \dot{q}\} = 0. \quad (11)$$

Equation (11) is of exactly the same form as equation (9) for the filament joint, however, it should be emphasized that the Lagrangians for the two models are completely different due to the different interpolation functions.

RESULTS

The results consist of strain-time plots for the four models (experiment, filament rigid block and triangular finite element). The experimental system consists of lap joint beams which are strain gauged at the positions shown in Fig. 1. It is of equal importance as the three theoretical models, and thus one cannot say that one theoretical model is better than the others, simply because it gives results closer to those obtained from the experimental system.

In all cases the theoretical models are compared against the experimental system since their predictive ability is being assessed and result from a triangular tensile pulse, Fig. 1.

In the right angled beam system it is seen that generally the reflected, longitudinal, shear and bending strains (beam 1) are well predicted (Fig. 11). The transmitted longitudinal strains are obviously small and one should not really try to compare these signals; the transmitted shear strains do have a general qualitative similarity between the models, but the quantitative dissimilarity is obviously dictated by, (i) the shear behaviour of the lapped joint and, (ii) the Timoshenko approximation of constant shear distribution over the beam width, whereas elementary strength of material theory predicts parabolic shear distributions.

For the 45° angled beam system only the filament joint model is considered since the conclusions made for the right angled system can be applied and the remaining results will not be included here in the interests of economy. The results are shown in Fig. 12 and it can be seen that the reflected signals are generally under-estimates, whereas the transmitted signals are over-estimates: this implies that the mechanical joint in the experimental system is not as effective as the theoretical system and that some improvements in the mechanical joints could be made.

Again, for the inclined cruciform system, only the filament joint model results are included, Fig. 13. In this case the only inaccurate predictions occur in shear, but these values are generally small and not important from the point of view of structural assessment.

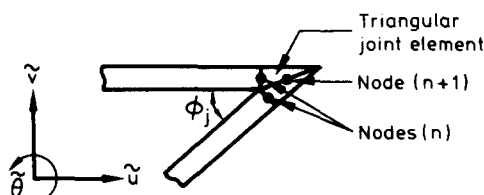


FIG. 9. Triangular element approximation.

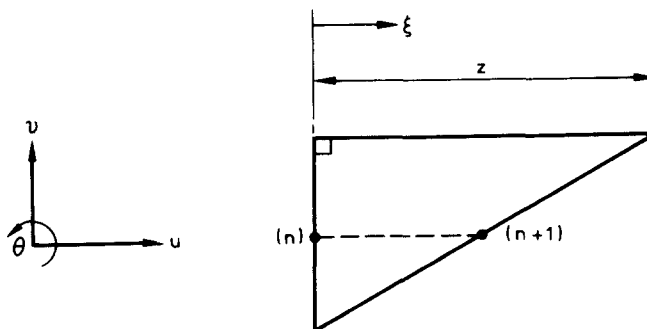


FIG. 10. Triangular element.

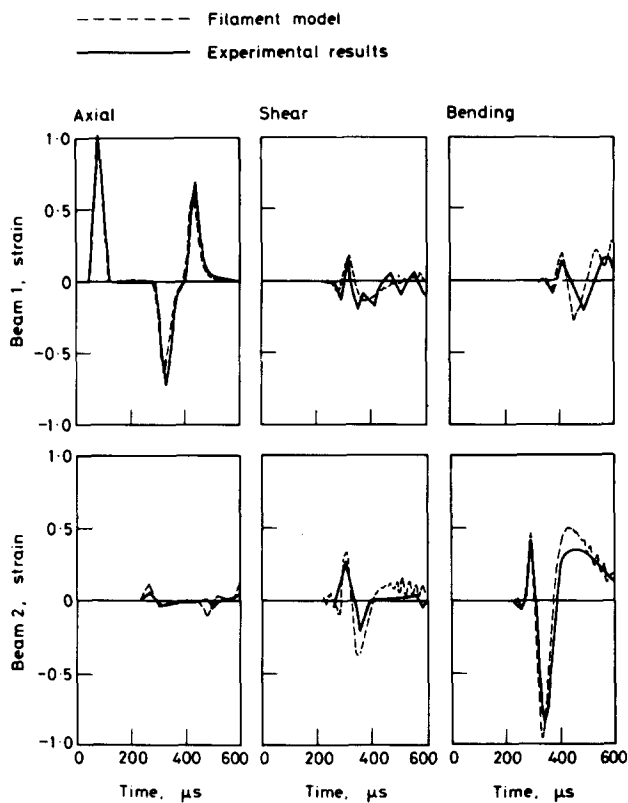


FIG. 11(a). Strain-time plots for right angled beam system—Filament joint model.

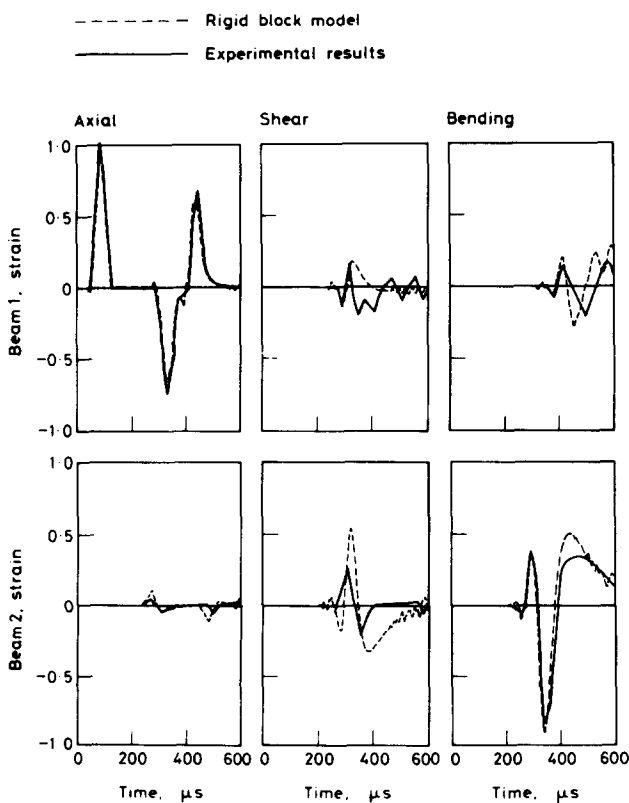


FIG. 11(b). Strain-time plots for right angled beam system—Rigid joint model.

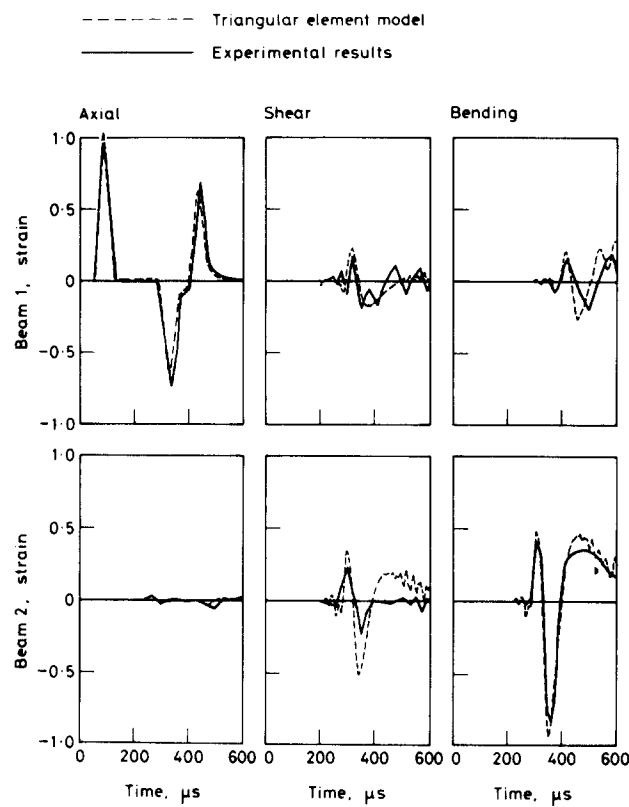


FIG. 11(c). Strain-time plots for right angled beam system—Triangular element joint model

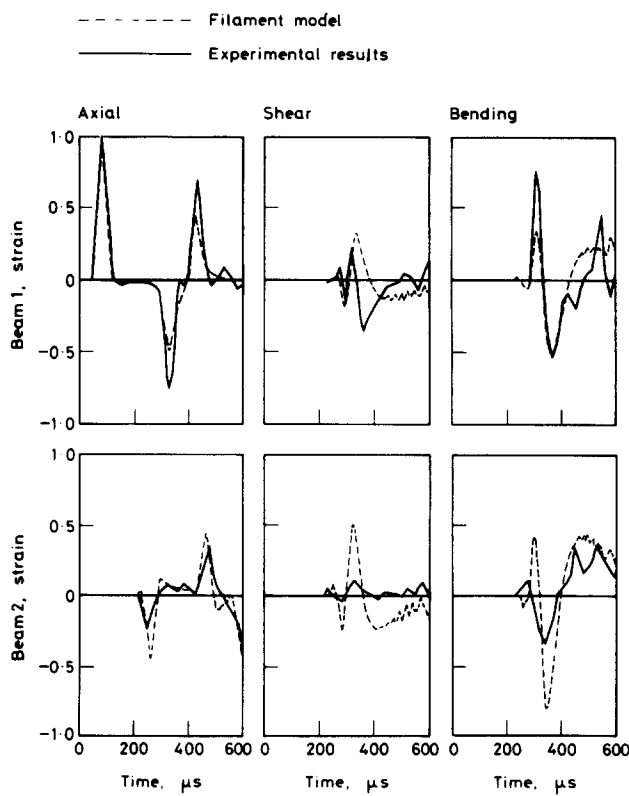


FIG. 12. Strain-time plots for 45° beam system—Filament joint model.

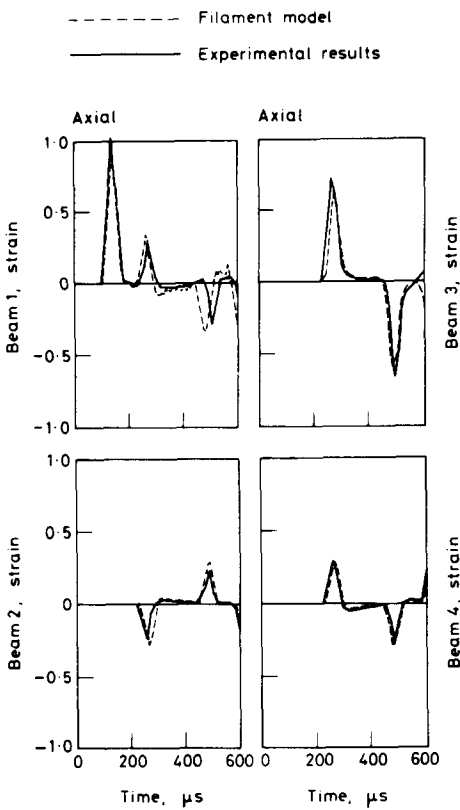


FIG. 13(a). Strain-time plots for angled cruciform beam system—Filament joint model.

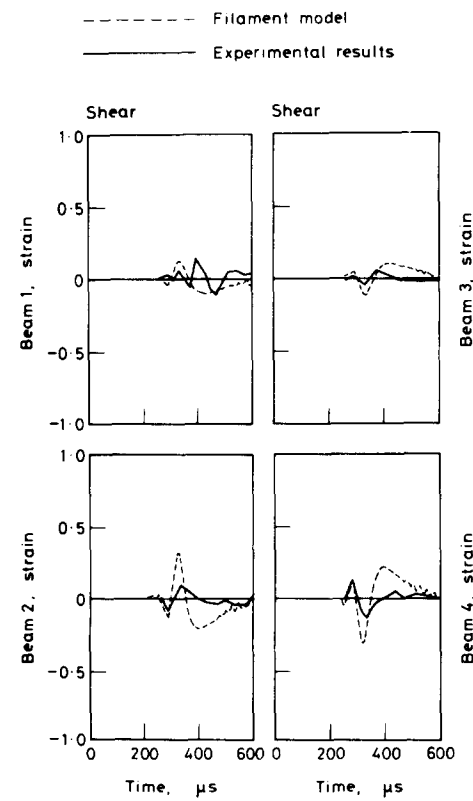


FIG. 13(b). Strain-time plots for angled cruciform beam system—Filament joint model.

Finally, pictorial or qualitative assessment of joint effectiveness can be made using the time plots, Figs. 14 and 15; These are very efficient methods of observing the propagation through joints and the reflection from the joint. Fig. 16 shows the motion and rotation of the joint based upon the unit tensile strain pulse. In each case it can be seen that the joint moves away from the branch, thus setting up in the branch a tensile axial strain wave for the right angle system and a reduced compressive strain wave in the 45° system.

Finally, it is pointed out that an extensional signal upon encountering a joint gives rise to reflected and transmission shear and flexural waves. This has the effect then of increasing the difference between maximum and minimum principal stresses and thus increasing the likelihood of fracture. This is indeed a subject for further investigation.

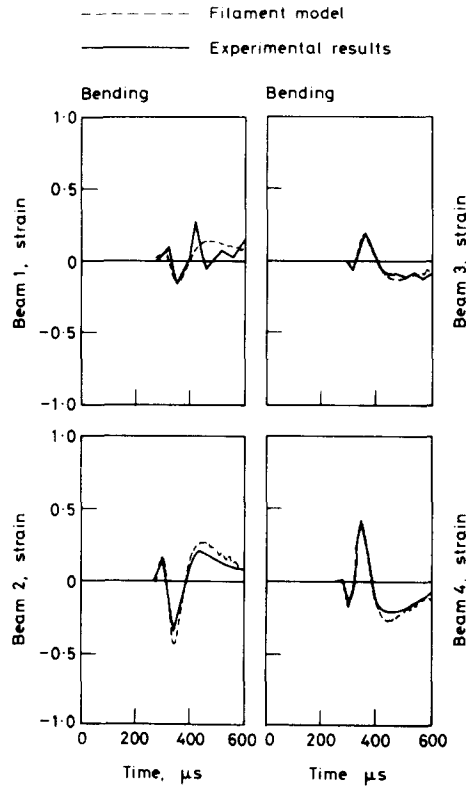


FIG. 13(c). Strain-time plots for angled cruciform beam system—Filament joint model.

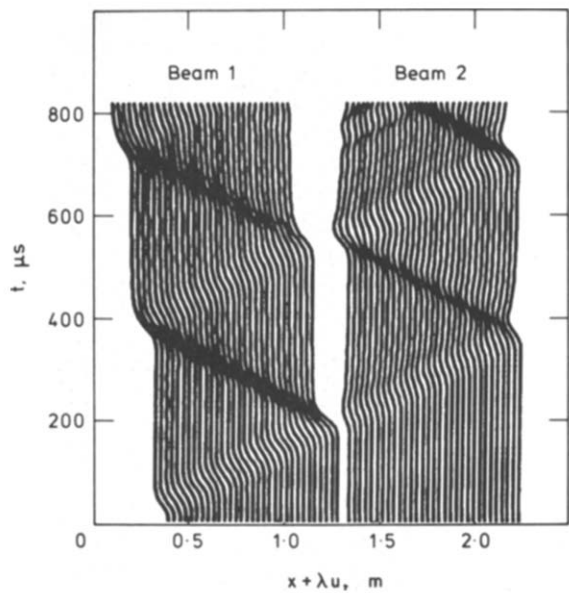


FIG. 14(a). Scaled axial displacements of right angled beam system—Filament joint model. (λ is chosen for pictorial effect to highlight stress waves.)

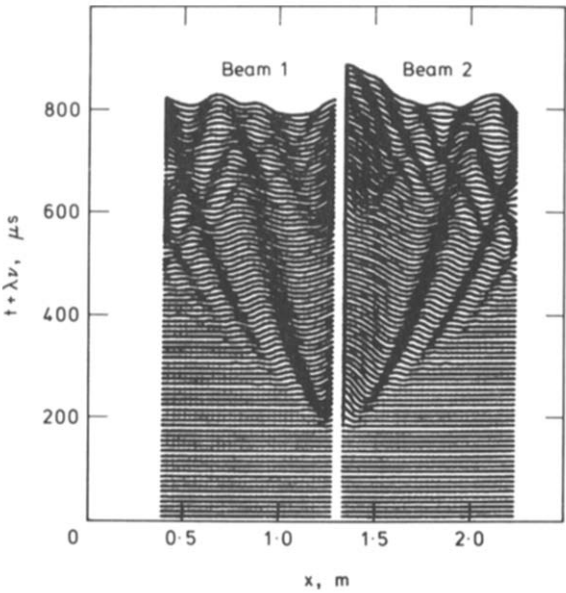


FIG. 14(b). Scaled transverse nodal displacements of right angled beam system—Filament joint model.

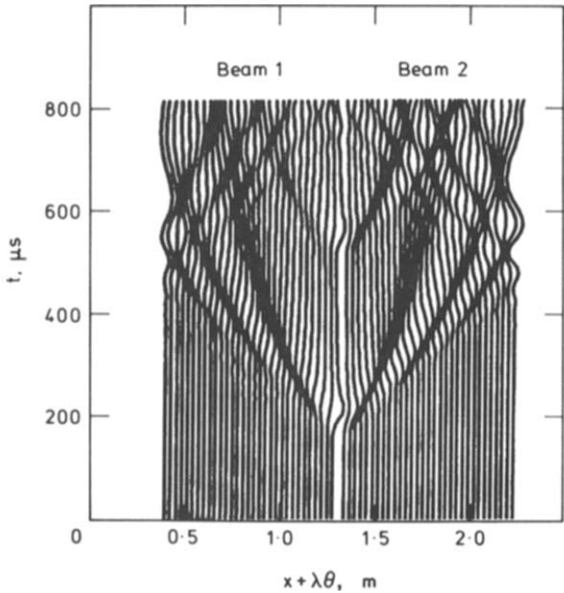


FIG. 14(c). Scaled nodal rotations of tight angled beam systems—Filament joint model.

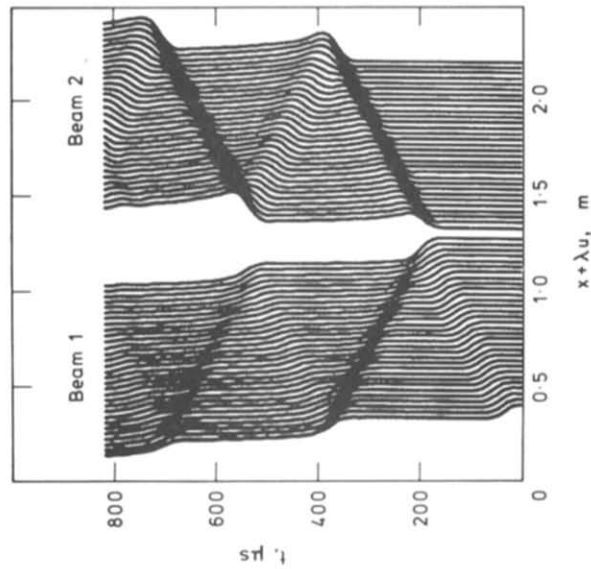


FIG. 15(a). Scaled axial displacements of 45° beam system—Filament joint model.

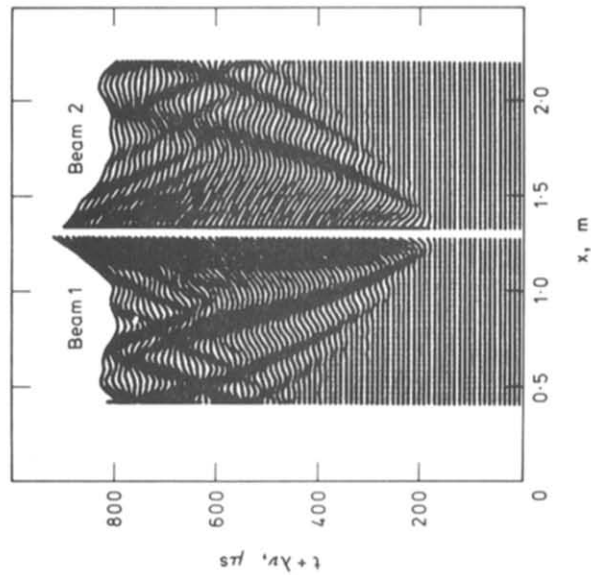


FIG. 15(b). Scaled transverse nodal displacements of 45° beam system—Filament joint model.

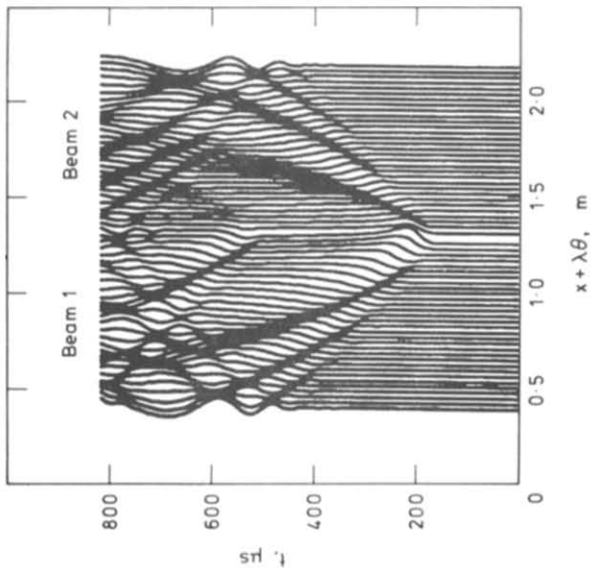


FIG. 15(c). Scaled nodal rotations of 45° beam system—Filament joint model.

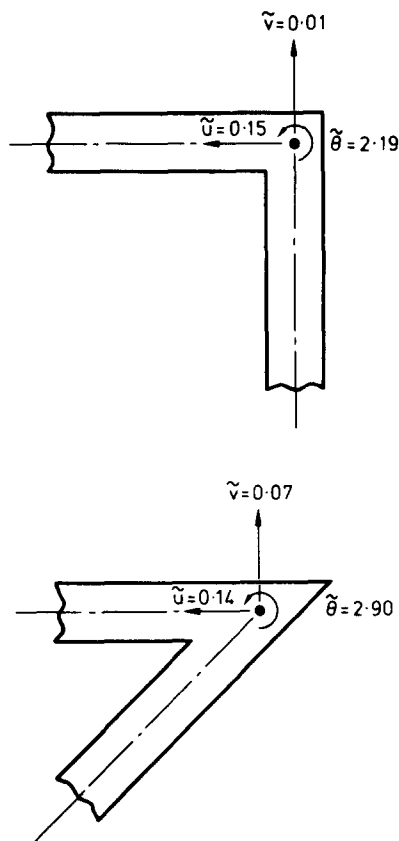


FIG. 16. Relative displacements at joint upon unit incident strain pulse—Filament joint model.

CONCLUSIONS

Three theoretical joints are considered for use in the modelling of the dynamics of jointed beam systems. These are used along with the Timoshenko beam and a finite element analysis. It has been found that there are no significant differences between the methods, although none predict accurately small transmitted shear signals. This is probably due to the Timoshenko approximation of constant shear through the section, the assumption not being valid through the joint.

It is especially significant that all the models predict quite accurately the large transmitted bending strains in the right angled beam system and in the inclined cruciform system.

Further development is needed for an improvement of shear prediction and for the combined flexure-torsion-extension dynamics of coupled beams.

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